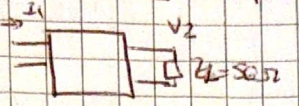
7) Sintetizar la transferencia no disipativa → LC

$$\left|\frac{V_2}{V_1}(j\omega)\right| = \sqrt{\frac{k^2}{1+\omega^6}}$$



⇓

$$\left|\frac{V_2}{V_1}(j\omega)\right|^2 = \frac{K^2}{1+\omega^6}$$ → Butter de tercer orden ($n=3$) como $A(s) = \frac{K}{s^3+2s^2+2s+1}$

- $V_2 = Z_{21} \cdot I_1 + Z_{22} \cdot I_2$
- $V_2 = Z_L(-I_2)$

$\Big\}$ $V_2 = Z_{21} I_1 + Z_{22}\left(-\frac{V_2}{Z_L}\right) \Rightarrow \frac{V_2}{I_1} = \frac{Z_{21}}{\frac{Z_L+Z_{22}}{Z_L}} = \frac{Z_{21} Z_L}{Z_L+Z_{22}}$

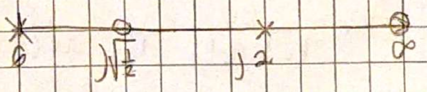
$V_2\left(1+\frac{Z_{22}}{Z_L}\right) = Z_{21} I_1$

→ Lo que podemos hacer es normalizar la RL ⇒ $Z_L = 1\Omega$ ⇒ $\frac{V_2}{I_1} = \frac{Z_{21}}{1+Z_{22}}$

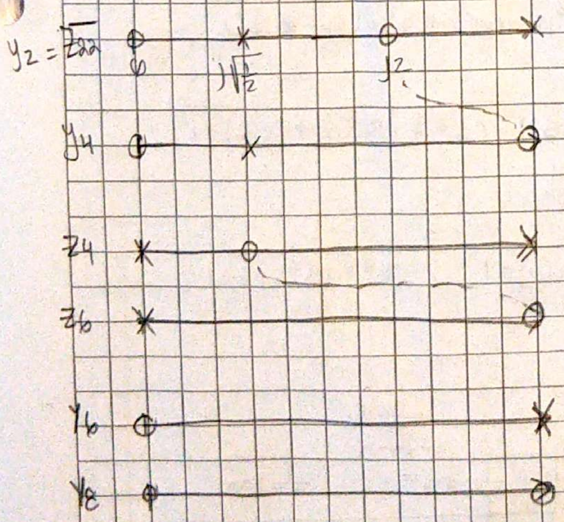
→ $\frac{V_2}{I_g} = \frac{1}{C'} = \frac{1}{\cancel{1/Z_G} \cdot A + C + \cancel{1/Z_G} \cdot Y_L \cdot B + Y_L D} = \frac{1}{Y_L \cdot D + C} \Rightarrow \frac{Z_{21}}{1+Z_{22}}$

→ $\frac{K}{s^3+2s^2+2s+1} = \frac{Z_{21}}{1+Z_{22}} \Rightarrow \frac{\frac{K}{s^3+2s}}{\frac{s^3+2s^2+2s+1}{s^3+2s}} = \frac{\frac{K}{s^3+2s}}{1+\frac{2s^2+1}{s^3+2s}}$ dividiendo por la parte impar!

$Z_{22} = \frac{2s^2+1}{s^3+2s}$ ← $s(s^2+2)$

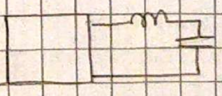


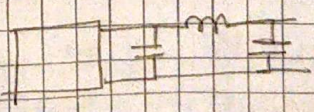
Buscamos conseguir los ceros de la función de transferencia



sacamos un polo en infinito.
↓
cap en derivación

lo mismo pero en impedancia





Como trabajamos con Z_{22}, empezamos desde la salida

$$y_2 = \frac{s^3+2s}{2s^2+1}$$

$\rightarrow y_4 = y_2 - K_\infty \cdot s$ con $K_\infty = \lim_{s\to\infty} \frac{y_2}{s} = \lim_{s\to\infty} \frac{s^3+2s}{(2s^2+1)s} = \frac{1}{2}$

$$y_2 = \frac{s^3+2s}{2s^2+1} - \frac{s}{2} = \frac{(s^3+2s)2 - s(2s^2+1)}{(2s^2+1)2} = \frac{2s^3+4s-(s^3 \cdot 2+s)}{2(s^2 \cdot 2+1)} = \frac{3s}{2(2s^2+1)}$$

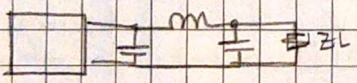
$\rightarrow Z_2 = \frac{2s^2+1}{\frac{3}{2}s} \Rightarrow Z_4 = Z_2 - K_\infty \cdot s$ con $K_\infty = \lim_{s\to\infty} \frac{Z_2}{s} = \lim_{s\to\infty} \frac{2s^2+1}{\frac{3}{2}s} \cdot \frac{1}{s} = \frac{4}{3}$

$$Z_4 = \frac{2s^2+1}{\frac{3}{2}s} - \frac{4}{3}s = \frac{3(2s^2+1) - 4s \cdot \frac{3}{2}s}{\frac{9}{2}s} = \frac{6s^2+3-6s^2}{\frac{9}{2}s} = \frac{3}{\frac{9}{2}s} = \frac{2}{3s}$$

$\rightarrow y_4 = \frac{3s}{2} \Rightarrow y_6 = y_4 - K_\infty \cdot s$ con $K_\infty = \lim_{s\to\infty} \frac{y_4}{s} = \lim_{s\to\infty} \frac{3s}{2s} = \frac{3}{2}$

$$y_6 = \frac{3s}{2} - \frac{3}{2}s = 0$$

Ahora con el circuito completo y recordando que está normalizado



y tenemos que desnormalizar con $Z_L = 50\Omega \wedge W_0 = 1 \text{ rad/seg}$

$N_Z = 50 \wedge N_\Omega = 1$

$$V_2 = \frac{Z_L \cdot Z_{C2}}{Z_L + Z_{C2}} \cdot \frac{Z_{C6}}{Z_{C6} + Z_{L4} + Z_L // Z_{C2}} \cdot I_1$$

$$V_2 = \frac{R_L \cdot \frac{1}{sC_2}}{R_L + \frac{1}{sC_2}} \cdot \frac{\frac{1}{sC_6}}{\frac{1}{sC_6} + sL_4 + R_L // \frac{1}{sC_2}} \cdot I_1$$

$$V_2 = \frac{R_L}{R_L \cdot sC_2 + 1} \cdot \frac{\frac{1}{sC_6}}{\frac{1}{sC_6} + sL_4 + \frac{R_L}{R_L sC_2 + 1}} \cdot I_1$$

$$V_2 = \frac{R_L}{R_L \cdot s \cdot C_2 + 1} \cdot \frac{1}{1 + s^2 L_4 C_6 + \frac{R_L \cdot sC_6}{R_L sC_2 + 1}} \cdot I_1$$

$$\frac{V_2}{I_1} = \frac{R_L}{sR_LC_2 + 1 + s^3 R_L C_2 L_4 C_6 + s^2 L_4 C_6 + R_L s C_6}$$

$$\frac{V_2}{I_1} = \frac{R_L}{s^3 R_L L_4 C_2 C_6 + s^2 L_4 C_6 + s(R_L C_2 + R_L C_6) + 1}$$

$$\Downarrow$$

$$\frac{R_L}{s^3 R_L L_4 C_2 C_6 + s^2 L_4 C_6 + s(R_L C_2 + R_L C_6) + 1} = \frac{1}{s^3 + 2s^2 + 2s + 1}$$

- $R_f = R \cdot N_Z = 1 \cdot 50\Omega = \boxed{50\Omega}$
- $L_{4f} = L_4 \cdot N_Z = \frac{4}{3} \cdot 50 = \boxed{\frac{200}{3} Hy}$
- $C_{2f} = \frac{C_2}{N_Z} = \frac{1}{2} \cdot \frac{1}{50} = \boxed{10mF}$
- $C_{6f} = \frac{C_6}{N_Z} = \frac{3}{2} \cdot \frac{1}{50} = \boxed{30mF}$